

OPTIMAL MONETARY POLICY WITH REDISTRIBUTION

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MOTIVATION

Recent calls for central banks to include tackling inequality into their mandates:

- Unequal incidence of business cycles (Parker and Vissing-Jorgenson, 2009; Guvenen et al. 2017; Alves et al., 2020; Patterson, 2023)
 - Inequality measures moving at the frequency of business cycle
 - Sticky fiscal policy
- Monetary policy has distributional impact (Coibion et al.; 2017; Auclert; 2019; Doepke and Schneider; 2006)
 - Through its effects on prices, monetary policy operates an “inflation tax”

This paper:

- asks (and answers analytically) how monetary policy can be used as a redistributive tool in the business cycles
- does not discuss heterogeneous effects of money shocks and focuses on the “inflation tax”

ABSTRACT

This paper studies optimal monetary policy in a dynamic, general equilibrium economy with

- Heterogeneous agents (cf. the HANK literature)
- Complete markets
- Uniform, state-contingent lump-sum tax, and state-independent linear taxes
- Firms facing information frictions in price setting

It shows that

- When the aggregate shock **does not change** the relative skill levels of households, all redistribution is done by fiscal policies. In this case it is optimal for monetary policy to **implement flexible-price** allocations
- On the other hand, when there are **shocks to the relative skill distribution**, optimal monetary policy implements a **state-contingent mark-up** that co-moves **positively** with a sufficient statistic for **labor income inequality**.

Intuition: MP targets higher mark-ups $\rightarrow$ lower real wages $\rightarrow$ more redistribution

ROADMAP

- 1 MODEL
- 2 EQUILIBRIUM DEFINITION AND ALLOCATIONS
- 3 THE RAMSEY PROBLEM
- 4 THE ROLE OF MONETARY POLICY

FRAMEWORK

- Discrete time
- Finite state space: $s_t \in S$, history $s^t = (s_0, \dots, s_t) \in S^t$
- Conditional probabilities $\mu(s_t | s^{t-1})$

Households

- Unit mass of households
- Identical preference:

$$U(c, h) = \frac{c^{1-\gamma}}{1-\gamma} - \frac{h^{1+\eta}}{1+\eta}$$

- Finite types $i \in I$ of relative size π^i
- types correspond to state-contingent worker skill

$$\theta^i(s_t) > 0$$

FRAMEWORK: HOUSEHOLDS

- efficiency units of labor:: $l^i(s^t) = \theta^i(s^t)h^i(s^t)$
- expected lifetime utility $\sum_t \sum_{s^t} \beta^t \mu(s^t) U(c^i(s^t), l^i(s^t)\theta^i(s^t))$
- subject to the budget constraint

$$(1 + \tau_c)P(s^t)c^i(s^t) + b^i(s^t) + \sum_{s^{t+1}|s^t} Q(s^{t+1}|s^t)z^i(s^{t+1}|s^t) \leq$$

$$(1 - \tau_l)W(s^t)l^i(s^t) + P(s^t)T(s^t) + (1 - \tau_\Pi)\Pi(s^t) + z^i(s^t|s^{t-1}) + (1 + i(s^{t-1}))b^i(s^{t-1})$$

- Assets: complete set of Arrow securities; nominal, one-period bonds (redundant)
- linear taxes on consumption, labor income, and profits
- lump-sum, state-contingent transfer $T(s^t)$
- uniform equity shares of firms

INTERMEDIATE GOOD FIRMS

- Continuum of intermediate good firms, indexed by $j \in \mathcal{J} = [0, 1]$ with technology

$$y^j(s^t) = A(s_t)n^j(s^t)$$

- Produce differentiated goods i.e. monopolistic competitors
- A share of $1 - \kappa$ firms $j \in \mathcal{J}^f$ are **attentive** and set price based on current history s^t :

$$p_t^j(s^t) \in \operatorname{argmax}_{p'} \left\{ (1 - \tau_r)p'y^j(s^t) - W(s^t)\frac{y^j(s^t)}{A(s_t)} \right\}$$

- A share of κ firms $j \in \mathcal{J}^s$ are inattentive and set price based on **last-period history** s^{t-1}

$$p_t^j(s^t) \in \operatorname{argmax}_{p'} \sum_{s^t|s^{t-1}} Q(s^t|s^{t-1}) \left\{ (1 - \tau_r)p'y^j(s^t) - W(s^t)\frac{y^j(s^t)}{A(s_t)} \right\}$$

A representative firm:

$$Y(s^t) = \left[\int_{j \in \mathcal{J}} y^j(s^t)^{\frac{\rho-1}{\rho}} dj \right]^{\frac{\rho}{\rho-1}}$$

- Perfectly competitive
- Maximizes profits

$$P(s^t)Y(s^t) - \int_{j \in \mathcal{J}} p_t^j y^j(s^t) dj$$

- Has the standard downward-sloping CES demand function for intermediate goods j :

$$y^j(s^t) = \left(\frac{p_t^j}{P(s^t)} \right)^{-\rho} Y(s^t)$$

THE GOVERNMENT

The government has a consolidated monetary and fiscal authority; it makes tax revenue

$$\mathcal{T}(s^t) = \tau_c P(s^t) C(s^t) + \tau_l W(s^t) L(s^t) + \tau_r P(s^t) Y(s^t) + \tau_\Pi \Pi(s^t)$$

The government has a budget constraint

$$\begin{aligned} P(s^t) T(s^t) + Z^i(s^t | s^{t-1}) + (1 + i(s^{t-1})) B^i(s^{t-1}) + P(s^t) G(s_t) \\ \leq B(s^t) + \sum_{s^{t+1} | s^t} Q(s^{t+1} | s^t) z^i(s^{t+1} | s^t) + \mathcal{T}(s^t) \end{aligned}$$

Where $G(s_t)$ is an exogenous government spending shock. Monetary authority controls money demand:

$$M(s^t) = P(s^t) C(s^t)$$

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EQUILIBRIUM DEFINITION

We're interested in a set of allocations

$$x = \{(c^i(s^t), l^i(s^t))_{i \in I}, (y^i(s^t), n^i(s^t))_{j \in J}, C(s^t), Y(s^t), L(s^t)\}_{t \geq 0, s^T \in S^t}$$

that is feasible (satisfies the economy's technology and resource constraints); a policy

$$\Omega = \{\tau_c, \tau_l, \tau_r, \tau_\pi, T(s^t), i(s^t)\}_{t \geq 0, s^T \in S^t};$$

a price system

$$\rho = \{p_t^f(s^t), p_t^s(s^{t-1}), P(s^t), W(s^t), Q(s^{t+1}|s^t)_{s^{t+1} \in S^{t+1}}\}_{t \geq 0, s^t \in S^t}$$

and a set of financial asset positions by

$$\zeta = \{(b^i(s^t))_{i \in I}, B(s^t), (z^i(s^{t+1}|s^t), Z(s^{t+1}))_{s^{t+1} \in S^{t+1}}\}_{t \geq 0, s^T \in S^t}$$

STICKY-PRICE EQUILIBRIUM

Define a **sticky-price equilibrium** and a benchmark **flexible-price equilibrium**:

DEFINITION

A sticky-price equilibrium is an allocation x , a price system ρ , a policy Ω , and asset holdings ζ such that: (i) at time t , history s^t , the price $p_t^s(s^{t1})$ is optimal for all sticky-price firms $j \in J^s$, the price $p_t^f(s^t)$ is optimal for all flexible-price firms $j \in J^f$, and the aggregate price level given by:

$$P(s^t) = \left[\kappa p_t^s(s^{t-1})^{1-\rho} + (1 - \kappa) p_t^f(s^t)^{1-\rho} \right]^{\frac{1}{1-\rho}};$$

(ii) prices and allocations satisfy the CES demand function for all $j \in J$ at time t ; (iii) given the price system and the policy, the allocation and financial asset holdings of type i solve the household problem of type i , for every $i \in I$; (iv) the government budget constraint is satisfied; (v) aggregate nominal demand satisfies $P(s^t)C(s^t) = M(s^t)$; and (vi) markets clear.

FLEXIBLE-PRICE EQUILIBRIUM

DEFINITION

A flexible-price equilibrium is a quadruple $\{x, \rho, \Omega, \zeta\}$, such that such that: (i) at time t , history s^t , the price $p_t^f(s^t)$ is optimal for all firms, and

$$P(s^t) = p_t^f;$$

and parts (ii)-(vi) of the previous definition.

THE PRIMAL APPROACH

- We'd like to solve for the optimal monetary policy using the primal approach
- Primal as in characterize the set of all feasible allocations that can be implemented as a competitive equilibrium with necessary and sufficient conditions that **only depend on the allocations**;

We'll end up with

- ① the resource constraint;
- ② the implementability condition: household FOCs plugged into budget constraints in place of prices and policies

The following result suggests that the economy behaves as if there is a representative agent with utility function $U^m(s^t)(C, L)$. Then we can express (real, after-tax) prices in terms of representative agent's allocations, and express individual household allocations as a share of the aggregates.

FICTITIOUS REPRESENTATIVE AGENT

LEMMA

(Werning, 2007) For any equilibrium there exist “market” weights $(\phi^i)_{i \in I}$ with $\phi^i \geq 0$ so that the individual assignments of consumption and labor solve the following static sub-problem

$$U^m(C(s^t), L(s^t); \phi) = \max_{(c^i(s^t), l^i(s^t))_{i \in I}} \sum_{i \in I} \phi^i \pi^i U(c^i(s^t), l^i(s^t)/\theta^i(s^t))$$

subject to

$$C(s^t) = \sum_{i \in I} \pi^i c^i(s^t), \text{ and } L(s^t) = \sum_{i \in I} \pi^i l^i(s^t)$$

where the superscript “m” stands for “market”. Then the solutions are:

$$c^i(s^t) = \omega_C^i(\phi) C(s^t), \text{ and } l^i(s^t) = \omega_L^i(\phi, s^t) L(s^t).$$

FLEXIBLE-PRICE EQUILIBRIUM

(Household's lifetime budget) implementability condition:

$$\sum_t \sum_{s^t} \beta^t \mu(s^t) [U_C^m(s^t) \omega_C^i(\phi) C(s^t) + U_L^m(s^t) \omega_L^i(\phi) L(s^t)] = U_C^m(s_0) \bar{T}$$

PROPOSITION

An implementable flexible-price allocation satisfies: for all $s^t \in S^t$,

$$-\frac{U_L^m(s^t)}{U_C^m(s^t)} = \chi A(s^t),$$

where $\chi = \left(\frac{\rho-1}{\rho}\right) \frac{(1-\tau_l)(1-\tau_r)}{1+\tau_c}$ (the fiscal policy).

- **Role of fiscal policy:** fiscal policy can control the labour wedge through χ (fixed).

PROPOSITION

An implementable sticky-price allocation satisfies: for all $s^t \in S^t$,

$$-\frac{U_L^m(s^t)}{U_C^m(s^t)} = \chi[\kappa\epsilon(s^t)^{1-\rho} + (1 - \kappa)]^{-\frac{1}{1-\rho}} A(s_t),$$

where $\epsilon(s^t)$ denotes a stochastic wedge between the optimal prices of the sticky- and flexible-price firms: $p^s(s^{t-1}) = \epsilon(s^t)p_t^f(s^t)$.

Power of monetary policy:

- Central bank can target $\epsilon(s^t) = 1$ no price prediction error (price stability), which replicates the flexible-price equilibrium.
- If set $\epsilon(s^t) \neq 1$, there's loss in production efficiency: $\frac{y^s(s^t)}{y^f(s^t)} = \left(\frac{p_t^s(s^t)}{p_t^f(s^t)}\right)^{-\rho} = \epsilon(s^t)^{-\rho}$

“INCORRECT” FISCAL POLICY

If the current fiscal policy χ is undesirable, can monetary policy make up for the mistakes?

- The paper argues no: because the pricing error $\epsilon(s^t)$ has to sum up to one ex ante, the monetary authority cannot target the same mark-up resulted from this error in all states
- The implicit timing here is that
 - 1 nature chooses state s_t ,
 - 2 the flexible firms learn it and set prices
 - 3 the sticky firms set prices without knowing the state
 - 4 state is revealed
- If the central bank learns the state before firms, then can potentially target an ϵ for the given state?

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WELFARE FUNCTION AND RAMSEY PROBLEM DEFINITION

Let $\mathcal{X}^f$ denote the set of all flexible-price equilibrium allocations and $\mathcal{X}^s$ sticky-price equilibrium allocations.

Consider a utilitarian planner with social welfare function given by:

$$\mathcal{U} = \sum_{i \in I} \lambda^i \pi^i \sum_t \sum_{s^t} \beta^t \mu(s^t) U(c^i(s^t), l^i(s^t) / \theta^i(s^t))$$

where λ^i denotes the Pareto weights.

DEFINITION

A Ramsey optimum x^* is an allocation x that maximise the above social welfare function subject to $x \in \mathcal{X}^s$.

THE RELAXED RAMSEY PROBLEM

Now we're ready to characterize the social welfare-maximizing allocations, and back out optimal fiscal/monetary policy. But let's tackle an easier problem first: enough for the constant skill ratio case

DEFINITION

The relaxed set of allocations $\mathcal{X}^R$ is the set of all feasible allocations $x \in X$ for which there exists a set of market weights (ϕ^i) such that the implementability condition

$$\sum_t \sum_{s^t} \beta^t \mu(s^t) [U_C^m(s^t) \omega_C^i(\phi) C(s^t) + U_L^m(s^t) \omega_L^i(\phi) L(s^t)] = U_C^m(s_0) \bar{T}$$

holds for all $i \in I$. A relaxed Ramsey optimum x^R is an allocation x that maximizes social welfare subject to $x \in \mathcal{X}^R$.

Relaxed: planner not restrained by prices

IMPLEMENT x^{R*} UNDER FLEXIBLE PRICE?

Why is the relaxed problem relevant?

LEMMA

Let $\mathcal{X}^f$ denote the set of flexible-price allocations, $\mathcal{X}^s$ the set of sticky-price allocations, and $\mathcal{X}$ all implementable allocations. Then $\mathcal{X}^f \subset \mathcal{X}^s \subset \mathcal{X}$.

So hopefully we can show $x^{R*} \in \mathcal{X}^f$ and hence $\mathcal{X}^f$? Here's a sufficient condition:

THEOREM

If there exist positive scalars $(\theta^1, \theta^2, \dots, \theta^I) \in \mathbb{R}_+^I$ and a function $\Theta : S \rightarrow \mathbb{R}_+$ such that $\forall s_t \in S$

$$\theta^i(s^t) = \vartheta^i \Theta(s^t),$$

then $x^{R} \in \mathcal{X}^f$, it follows that*

$$x^{R*} \in \mathcal{X}^s$$

IDEA OF THE PROOF

- Relaxed Ramsey optimality condition

$$-\frac{\mathcal{W}_L(s^t)}{\mathcal{W}_C(s^t)} = A(s_t)$$

can be written as

$$-\frac{U_L^m(s^t)}{U_C^m(s^t)} \left[\frac{\sum_{i \in I} \pi^i \omega_L^i(\phi) \left(\frac{\lambda^i}{\phi^i} + v^i(1 + \eta) \right)}{\sum_{i \in I} \pi^i \omega_C^i(\phi) \left(\frac{\lambda^i}{\phi^i} + v^i(1 + \eta) \right)} \right] = A(s_t)$$

- Notice that the part in the square brackets doesn't depend on $s_t \rightarrow$ can be implemented as flexible-price allocation by linear taxation, while monetary policy targets price stability

INTUITION

- Think about the trade-off of the Ramsey planner in a steady state
- Cost: production efficiency, which is constant due to homothetic technology and preference
- Benefit: equality in after-tax real wage

$$-\frac{U_h^i}{U_c^i} = -\frac{U_l^i \theta^i}{U_c^i} = \frac{U_L^m}{U_C^m} \frac{\theta^i \omega_l^i}{\omega_c^i} = \chi^A \frac{\theta^i \omega_l^i}{\omega_c^i}$$

- In face of a shock: constant relative skill distribution, constant relative real-after-tax wage distribution:

$$\frac{U_h^k / U_c^k}{U_h^i / U_c^i} = \frac{\omega_c^i}{\theta^i \omega_l^i} \frac{\theta^k \omega_l^k}{\omega_c^k}$$

- Tools for redistribution: no state-contingent redistribution acquired, monetary policy targets price stability for constant labour wedge; fiscal policy as usual

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THE UNRELAXED RAMSEY PROBLEM

- What if the sufficient condition does not apply, i.e. $x^{R*} \notin \chi^s$?

PROPOSITION

A Ramsey optimum x^* satisfies

$$\frac{-\mathcal{W}_L(s^t) + U_L^m(s^t)(\eta + 1) \left[\kappa \nu(s^{t-1}) \frac{y^s(s^t)}{A(s^t)L(s^t)} + (1 - \kappa) \xi(s^t) \frac{y^f(s^t)}{A(s^t)L(s^t)} \right]}{\mathcal{W}_C(s^t) + U_C^m(s^t)(\gamma + 1) \left[\kappa \nu(s^{t-1}) \left(\frac{y^s(s^t)}{Y(s^t)} \right)^{\frac{\rho-1}{\rho}} + (1 - \kappa) \xi(s^t) \left(\frac{y^f(s^t)}{Y(s^t)} \right)^{\frac{\rho-1}{\rho}} \right]} = \frac{Y(s^t)}{L(s^t)}$$

- That is a mess, but focus on the labour wedge -
- Define the implicit "monetary wedge" $1 - \tau_M^*(s^t)$ by

$$-\frac{U_L^m(s^t)}{U_C^m(s^t)} = \chi^*(1 - \tau_M^*(s^t)) \frac{Y(s^t)}{L(s^t)},$$

where χ^* is the implicit fiscal wedge.

A SUFFICIENT STATISTIC FOR THE LABOUR INCOME DISTRIBUTION

Let $\mathcal{I}:S \rightarrow \mathbb{R}_+$ be a positively-valued function defined by:

$$\mathcal{I}(s^t) = \frac{\sum_{i \in I} \tilde{\pi}_i (\phi^i)^{-1/\eta} (\theta^i(s^t))^{\frac{1+\eta}{\eta}}}{\sum_{i \in I} \pi_i (\phi^i)^{-1/\eta} (\theta^i(s^t))^{\frac{1+\eta}{\eta}}} > 0, \text{ where } \tilde{\pi}^i = \pi \left[\frac{\lambda^i}{\phi^i} + v^i(1 + \eta) \right]$$

- $\mathcal{I}(s^t)$ is monotonically increasing with regard to the “spread” of $\theta^i(s^t)$ s
- Is a sufficient statistic for the level of labour income inequality

THEOREM

There exists a threshold $\bar{\mathcal{I}}(s^t) > 0$, measureable in history s^{t-1} , such that optimal implicit monetary tax $\tau_M^(s^t)$ satisfies:*

$$\tau_M^*(s^t) > 0$$

if and only if $\mathcal{I}(s^t) > \bar{\mathcal{I}}(s^t)$

$$\tau_M^*(s^t) = 0$$

if and only if $\mathcal{I}(s^t) = \bar{\mathcal{I}}(s^t)$

$$\tau_M^*(s^t) < 0$$

if and only if $\mathcal{I}(s^t) < \bar{\mathcal{I}}(s^t)$

THEOREM

Define $\log \mathcal{M}(s^t) = \log P(s^t) - \log(W(s^t)/A(s^t))$, the log mark-up. Then

$\mathcal{M}(s^t) > 0$	<i>if and only if $\mathcal{I}(s^t) > \bar{\mathcal{I}}(s^t)$</i>
$\mathcal{M}(s^t) = 0$	<i>if and only if $\mathcal{I}(s^t) = \bar{\mathcal{I}}(s^t)$</i>
$\mathcal{M}(s^t) < 0$	<i>if and only if $\mathcal{I}(s^t) < \bar{\mathcal{I}}(s^t)$</i>

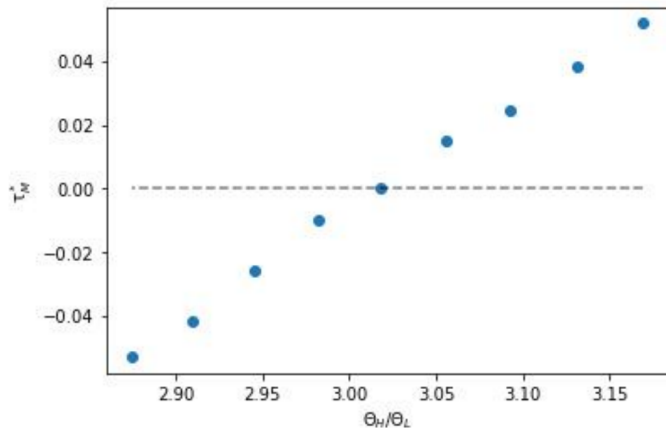


Figure 1. The optimal monetary tax $\tau_M^*(s^t)$ as a function of $\theta^H(s_t)/\theta^L(s_t)$

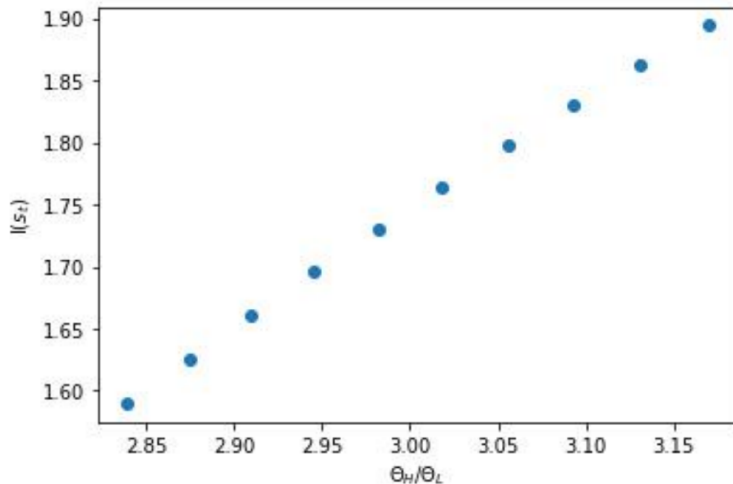


Figure 2. The inequality statistic $\mathcal{I}(s_t)$ as a function of $\theta^H(s_t)/\theta^L(s_t)$

- The previous theorem is for when the inequality is fixed at threshold level
- However, when relative skills are more dispersed -
- for example, let $\theta^i < \theta^k$, recall

$$\frac{U_h^k / U_c^k}{U_h^i / U_c^i} = \frac{\omega_c^i}{\theta^i \omega_l^i} \frac{\theta^k \omega_l^k}{\omega_c^k}$$

when θ^k / θ^i increases (high type even more productive), this ratio increases (more inequality) - more room for redistribution

- So, when labour income quality is strictly above a threshold, it's optimal to the increase level of redistribution...
- but the tax rates are fixed. So monetary authority mimics a “monetary tax” to fill the role
- Which is a wedge between sticky and flexible prices.

MONOTONICITY AND PARTIAL STATE-CONTINGENCY IN TAXES

Allow tax rates to be set one period in advance: $\tau_c(s^{t-1})$, $\tau_l(s^{t-1})$, $\tau_r(s^{t-1})$; then the optimal monetary policy is monotonically increasing in $\mathcal{I}(s_t)$.

THEOREM

Let tax rates be set one period in advance: $\chi(s^{t-1})$ is measurable in s^{t-1} . Then

- 1 $\tau_M^*(s^t)$ is strictly increasing in $\mathcal{I}(s_t)$,
- 2 there exists a threshold $\tilde{\mathcal{I}}(s^{t-1}) > 0$ such that $\tau_M^*(s^t) = 0$ if and only if $\mathcal{I}(s_t) = \mathcal{I}(s^{t-1})$, and
- 3 the derivative of $\tau_M^*(s^t) = 0$ with respect to $\mathcal{I}(s_t)$ at the threshold is given by:

$$\delta_0 = \left. \frac{d\tau_M^*(s^t)}{d\mathcal{I}(s_t)} \right|_{\mathcal{I}(s_t)=\tilde{\mathcal{I}}(s^{t-1})} = (1 - \kappa)(\eta + \gamma) + \frac{\kappa}{\rho} > 0$$

and

$$\frac{d\delta_0}{dp} < 0.$$

HETEROGENEOUS EQUITY SHARES

Let $1 + \sigma^i$ denote the share of equity held by a household of type i , with $\sum_{i \in I} \pi^i \sigma^i = 0$; equity shares are fixed.

THEOREM

Let tax rates be set one period in advance. Then:

- 1 there exists a threshold $\bar{\mathcal{I}}^\sigma(s^{t-1}) > 0$ such that $\tau_M^*(s^t) = 0$ if and only if $\mathcal{I}(s_t) = \bar{\mathcal{I}}^\sigma(s^{t-1})$,
- 2 the derivative at $\tau_M^*(s^t) = 0$

$$\delta_0 = \left. \frac{d\tau_M^*(s^t)}{d\mathcal{I}(s_t)} \right|_{\mathcal{I}(s_t) = \bar{\mathcal{I}}(s^{t-1})} > 0$$

- 3 the threshold $\bar{\mathcal{I}}^\sigma(s^{t-1})$ is increasing in $\sum_{i \in I} \pi^i \nu^i \sigma^i = 0$.

- $\sum_{i \in I} \pi^i \nu^i \sigma^i = 0$. is a measure of cross-sectional covariance between equity share and labour skill type (summarized by ν^i , which is the Lagrangian multiplier of budget implementability of type i);
- When this variance is high, the monetary authority finds it optimal to raise the threshold for the “monetary wedge”, as in, to become more tolerant towards labour income inequality
- High monetary tax in this case increase dividends pay-out inequality

CONCLUSION

- Discusses the redistribution role of monetary policy
- Monetary policy can be used as a state-contingent tax on the real wage through prices, at the cost of production efficiency
- Under equal incidence of business cycles, it's optimal for monetary policy to conduct the traditional role of stabilizing prices;
- when labour skill ratios change throughout the business cycle, the monetary policy should target a higher mark-up in face of higher inequality.